

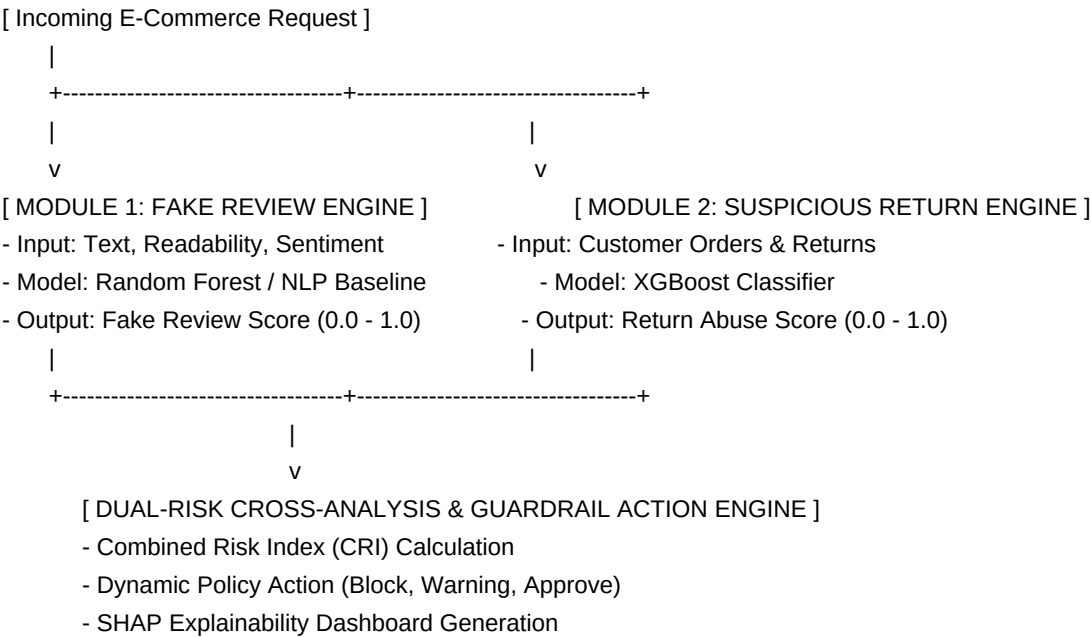
PART 1: PROJECT ARCHITECTURE & EXECUTION FLOW

1. OVERVIEW & SCOPE

This document outlines the complete execution flow for the project titled:
"E-Commerce Guardrail: Fake Review Detection and Suspicious Return Behaviour Prediction".
The goal is to build a dual-engine security framework using Machine Learning, NLP, and Explainable AI.

2. CORE SYSTEM ARCHITECTURE

The system operates as a dual-pipeline decision engine:



3. STEP-BY-STEP IMPLEMENTATION FLOW

- Step 1: Data Preparation & Preprocessing
 - * Master Review Dataset (reviews.csv, sentiment_labels.csv, products.csv, sellers.csv)
 - * Online Retail Dataset (online_retail_II.xlsx -> Customer Level Aggregation)
- Step 2: Feature Engineering
 - * Module 1: Sentiment Polarity, Subjectivity, Sarcasm flags, All-Caps Ratio, Readability Score.
 - * Module 2: Order Velocity, Return Ratio (returns/orders), Refund Ratio, Customer Lifetime Value.
- Step 3: Model Training & Persistence
 - * Train Random Forest Classifier for Fake Reviews (save as fake_review_model.pkl).
 - * Train XGBoost Classifier for Return Fraud (save as return_model.pkl).
- Step 4: Streamlit Dashboard Integration
 - * Build interactive multi-tab dashboard (app.py) incorporating live testing benches.